DAA EXPERIMENT 3

Bubble Sort with Big-O Analysis

Aim

To implement Bubble Sort using Python and analyze its time and space complexity.

Algorithm

1. Start.
2. Read the number of elements and store them in an array.
3. Compare each pair of adjacent elements.
4. If the first element is greater than the second, swap them.
5. Repeat the passes until the array is sorted.
6. Display the sorted array and analyze its complexity.
7. Stop.

Program

```
# Bubble Sort

n = int(input("Enter number of elements: "))
arr = []

for i in range(n):
    arr.append(int(input("Enter element: ")))

for i in range(n):
    for j in range(n - i - 1):
        if arr[j] > arr[j + 1]:
            arr[j], arr[j + 1] = arr[j + 1], arr[j]

print("Sorted array:", arr)
```

Complexity Analysis

- Best Case: $O(n^2)$
- Average Case: $O(n^2)$
- Worst Case: $O(n^2)$
- Space Complexity: $O(1)$

Output

```
Enter number of elements: 5
Enter element: 50
Enter element: 20
Enter element: 40
Enter element: 10
Enter element: 30
Sorted array: [10, 20, 30, 40, 50]
PS C:\Users\jithe\OneDrive\Documents\DAA - programs>
```

Result

Thus, Bubble Sort was successfully implemented and its time and space complexity were analyzed.